

# Experimental study of innovative periodic cellular structures as air volumetric absorbers

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## ABSTRACT

This work presents an experimental thermal evaluation of new and innovative porous morphologies suitable for volumetric solar receivers. Five new morphologies, four of them based on the repetition of single unit cells: diamond, rotated cube, tetrakaidecahedron, cube shifted, and a random morphology following the Voronoi tessellation technique were tested. In addition, for emerging geometries, two different cell size configurations were evaluated: one with constant cell size dimension and one with cell's size changing with flux direction. All the lattices were analysed experimentally in a lab-scale solar simulator. The results were compared with the ones of the already established and more conventional silicon carbide honeycomb and random foams proving that a gain in performance is achievable with the Voronoi morphology, both with constant cell size and increasing cell size pattern.

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## 1. Introduction

Porous structures are widely used for different industrial applications [1] such as thermal insulation [2], heat exchangers [3], industrial burners [4], thermal energy storage [5] and volumetric receivers [6] due to their interesting properties [7]. The application of porous structures together with the potential of solar thermal power tower plants emerge [8,9], firstly, to partially solve the high thermal losses and inefficient transmission of tubular receivers [10], and secondly, to address some of the concerns related to concentrated solar power like the cost of electricity and overall plant efficiency [11]. As a result, open volumetric receiver technology is considered a potential candidate due to its advantages [12], and its capacity to work at higher incident solar flux profiles, as formerly demonstrated by Fricker [13] and Hoffschmidt [14], which leads to the possibility of having higher working temperatures for the fluid [15].

Research on volumetric receivers started in the 80s, and for two decades a lot of prototypes were tested, mainly on an experimental

basis at the Plataforma Solar de Almería (PSA) due to its unique facilities [16–20]. Since then, most of the research work has been focused on the numerical analysis [21–25] of volumetric performance in order to maximize the conversion of the striking concentrated solar energy over the absorber into process heat that could be used later in power cycles of high efficiency [26,27].

The feasibility of volumetric absorbers using air depends on the capacity of the porous structure to convert the solar energy into thermal power with high thermal efficiency, and, without having structural damage at any possible air outlet temperature or compromising absorber integrity as a result of hot spots that could melt it. Thus, ceramic materials are the most widely studied ones in both experimental and numerical studies [28], having the HiTRec II structure [29] as the reference ceramic design. Nonetheless, several different absorber concepts and arrangements have been developed on the experimental level in order to meet the required conditions and the envisaged volumetric effect, which means the fluid temperature at the receiver outlet is higher than the porous temperature at the frontal surface [30,31]. The experiments in the literature that have come closest to producing the volumetric effect are, firstly, the study of Menigault [32] with a slab selective absorber, having a first layer made of glass-beads transparent to solar radiation but absorbent in the infrared range, and a second

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layer absorbent in both spectra, and secondly, the recent work of Luque [33] demonstrating that the theoretical benefit of a volumetric effect over a novel additively manufactured volumetric absorber was not observed. So, a review of the most interesting work during the last five years on the topic of experimental tests with open volumetric receiver is presented.

Mey et al. [34] developed a new device to test open volumetric receivers together with a homogenizer to have quasi-homogeneous concentrated solar radiation at the sample aperture. Seven different foam samples were tested, with porosities in the range of 72%–92% and pores per inch (PPI) varying between 5 and 20, together with a manufactured foam based on a new selective material, Zirconium Diboride. The results were compared to a silicon carbide honeycomb absorber, and the work concluded that, the foam with the smallest pore diameter and the lowest porosity produces the best results in terms of thermal efficiency. It is highlighted that the selective material results were promising. Capuano [35] presents a new demonstrator with a particular pin-shaped inlet zone coupled with an inner staggered-honeycomb geometry. This patented design was produced by a 3D printing technique with titanium-aluminium alloy and was tested experimentally in a laboratory test-bed. The experimental results were compared with a detailed simulation of the new geometry, and it was shown that this design was able to increase thermal efficiency by 12% compared to the HiTRec baseline ceramic absorber [29]. Pabst et al. [36] analysed a metal structure with a morphological structure similar to honeycombs, but having local variations in order to avoid a linear direction of the flow. Firstly, nine samples were tested experimentally in laboratory conditions with a solar simulator, and the results were analysed and compared with the numerical predictions, and then the three best samples were scaled-up to be tested in the Solar Tower of Jülich with a total of 54 cups, 18 cups per sample. The results were very promising, having 800 °C air outlet temperatures with thermal efficiencies around 80%, improving the performance of the main receiver in Jülich by 5%. Avila-Marin [37] focused on metallic wire mesh screens due to their commercial availability, easy manufacture and light weight. With this concept, 26 different configurations were tested experimentally; 6-absorbers with single porosity, 12-absorbers with double decreasing porosity and 8-absorbers with triple decreasing porosity. The results showed that 6 different configurations were able to improve on reference ceramic and metallic volumetric receivers' performance. Zaversky and co-workers [38] studied the performance of ceramic foams numerically and experimentally. The numerical results of two different models, were compared with experimental screening tests. During the experimental work, single-, double- and triple-porosity volumetric receiver configurations were tested, and both models agreed with the experimental findings. The best performance was achieved with the single porosity foams and the smallest strut diameter. Du [39] used a 3D printing technique to fabricate metallic foams with constant pore size and radial varied pore size, in order to compare both designs under heterogeneous concentrated solar flux. Experimental results with uniform porous receivers suggest that those with a smaller pore diameter perform the best, while the radial configurations with larger pores at the centre of the sample and smaller ones at the outer parts could further improve the thermal efficiency under non-homogeneous solar flux. Savoldi [40] analyses the applicability of using Raschig Rings as a porous media given their capacity to remove heat in the gyrotron resonant cavity in the nuclear fusion field. The concept was tested experimentally in the 60 kW solar furnace of PSA, checking the thermal and hydraulic performance under different solar fluxes and air mass flow rates. The results of the first campaign were very promising and another test campaign is planned in order to get to the operating limits by increasing the

solar flux over the design. Wang [41,42] studied experimentally and numerically the thermal and hydrodynamic behaviour of several silicon carbide (SiC) foams with four different porosities (65–70–75–85%) and three different pore sizes (15–20–40 PPI), showing that the sample with the largest specific surface area ( $1328 \text{ m}^{-1}$ ) performed the best. This sample has a porosity of 66% and 40 PPI. Patil et al. [43] report that the design and experimental tests of a cavity volumetric receiver can be analysed according to four different components: the porous material (silicon-infused silicon carbide, alumina, and ceria), the mean pore size (10–20–30 PPI, at 0.90 porosity), the concentration ratio (1965–3900 suns) and the air mass flow rate (2–10 kg/h). To conclude, several designs, with the proper selection of the different components, were able to exceed the envisaged 1000 °C with a reasonable thermal efficiency of around 65%.

For the first time, this work analyses experimentally the thermal performance of four new morphological configurations based on the replication of single unit cells (diamond, tetrakaidecahedron, rotated cube, cube shifted) and, two non-periodic structure (foam and Voronoi) for its application as volumetric receivers in an open volumetric receiver facility, and then, their results were compared with classical unstructured random foams [44] and structured honeycomb absorbers [45]. Moreover, gradual configurations, increasing and decreasing cell size with the fluid flow direction, were tested experimentally as well.

The article is structured as follows: In section 2, the fabrication process of the samples, their geometrical parameters and characteristics are described; section 3 presents the experimental methodology; in section 4, the experimental results are presented and discussed; finally, section 5 presents the conclusions of this work.

## 2. Innovative open volumetric absorber morphologies

### 2.1. Manufacturing process

The manufacturing process starts with design. For periodic cellular structures, a unit cell is selected, usually derived from crystal structures and eventually modified for specific desired end results. Next, a frame component is defined (boundary conditions), which reflects the overall dimensions (x,y,z) of the desired product. The volume (x,y,z-values) is scaled based on production shrinkage/expansion in an iterative design-prototyping process in order to allow near net shape manufacturing. The volume is then populated with the unit cells by repeating the x, y and z axes. For non-periodic structures, eg. Voronoi, a design technique called Voronoi tessellation is used. For this process, the defined frame volume is first filled with points which are subsequently connected to each other following user defined rules about the connectivity of each point. One of the interesting properties of Voronoi structures is the ability to insert porosity gradients within the structure. This has been exploited to produce structures with the desired geometrical features.

The designed structures are then printed through additive manufacturing of polymer [46]. The printed polymeric models serve as template structures for the ceramic shaping process [47], and give the desired morphology to the final prototype. Fig. 1 shows the schema of the manufacturing process followed for the morphologies.

The shaping of the ceramic item is done through a replica process known as the Schwarzwald method, which consists of coating a sacrificial polymeric template with a SiC-based ceramic slurry. The coated template structure, also called the green body, is then dried and pyrolyzed. A cycle with slow heating ramp up to 1000 °C in a flowing nitrogen atmosphere is elaborated to carefully pyrolyze additives and the polymeric template without the

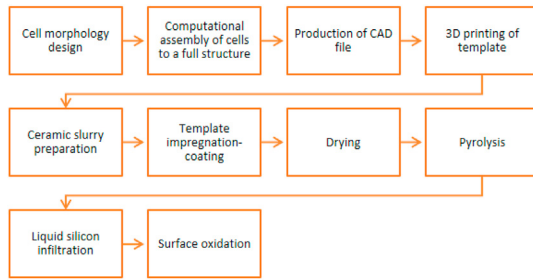


Fig. 1. Manufacturing process schema for SiSiC morphologies.

formation of cracks from thermal expansion or volume changes from degradation reactions. All organic additives are pyrolyzed and leave only amorphous carbon in the material matrix. Thus, at the end of the debinding cycle, a C/SiC preform is obtained (the brown body).

The brown body is then processed through reactive melt infiltration with silicon at 1500 °C in a vacuum furnace. The purpose of the process is to react carbon with silicon to form silicon carbide and to fill the residual porosity with free (unreacted) silicon to obtain a fully dense SiSiC microstructure.

The last step of the production is an oxidation cycle performed in air up to 1500 °C. The cycle serves to oxidize the surface of the lattices, reducing the reflected radiation. Fig. 2 depicts the production process with pictures of real materials.

## 2.2. Description of the morphologies

With the aim of identifying innovative open volumetric absorbers, new periodic and non-periodic morphologies are identified, manufactured, tested and compared with a reference structured honeycomb absorber [48,49] and with a typical unstructured random foam [50,51].

For this work, five innovative and novel morphologies, four periodic based on the repetition of single unit cells and one non-periodic, were manufactured: i) Diamond cell (D); ii) Tetrakaidecahedron cell (T); iii) Rotated Cube cell (RC); iv) Cube Shifted cell (CS); v) Voronoi (V), together with the more conventional random Foam (F) which are presented in Fig. 3. The first four units' cells are able to form an absorber based on the repetition of the single unit, while the last two are random foams manufactured with different tessellation techniques.

Concerning the state of the art of ceramics for its application in concentrated solar thermal power, two conventional morphologies are included in the study; the well-known and periodical honeycomb structure (H) available from previous tests, and the conventional random foam (F) manufactured for this work as well. Fig. 4

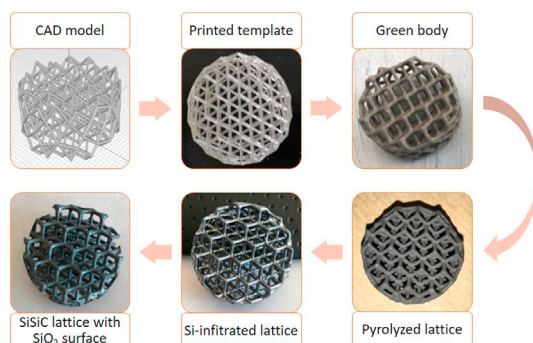


Fig. 2. Real pictures of the production process schema for SiSiC morphologies.

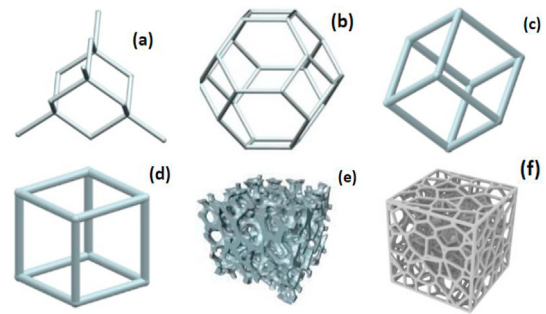


Fig. 3. Morphological cell units: a) Diamond, b) Tetrakaidecahedron, c) Rotated Cube, d) Cube Shifted, e) Foam, f) Voronoi.

shows the different pieces used as volumetric solar absorbers.

Diamond, random foam, tetrakaidecahedron and rotated cube lattices were manufactured with three different cell size (big-b, medium-m, small-s), the Voronoi were processed following two different arrangements, constant-c cell size and gradual-g cell size within the structure, and the cube shifted lattice was manufactured with constant shifted pores. The most significant properties of each lattice are presented in the next section.

Due to manufacturing constraints, the experimental tests were carried out in two different campaigns. Table 1 presents the lattices tested in each one. Firstly, all the lattices were tested, except the Voronoi morphologies. Later on, the Voronoi were manufactured and tested in a second campaign, in order to improve on the results of the first campaign.

### 2.2.1. First campaign

Among all the available samples with different pore sizes (D-T-RC-F), the small (s) pore was chosen as the most representative morphology for all the lattices ( $D_s$ - $T_s$ - $RC_s$ - $F_s$ ), as it has been demonstrated that higher efficiency is usually achieved for those specimens with smaller pores [34]. Moreover, the analysis of gradual configurations, increasing ( $D_{smb}$ - $RC_{smb}$ ) and decreasing ( $D_{bms}$ - $T_{bms}$ - $RC_{bms}$ - $F_{bms}$ ) cell size with the air flow direction, was considered and analysed experimentally.

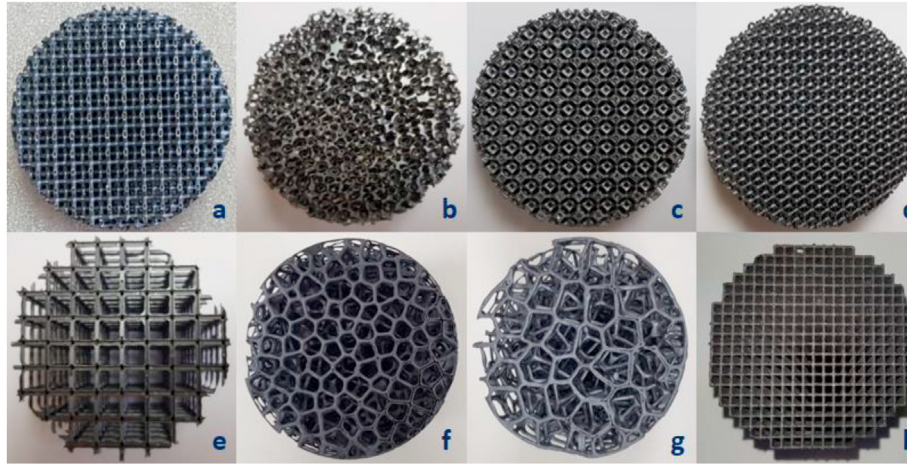
### 2.2.2. Second campaign

During the first test campaign, the preliminary results depicted the main conclusions of the campaign that were confirmed later, once the campaign ended. As a result, a new innovative morphology, Voronoi, was proposed. With such a morphology, two different patterns were adopted, one with constant pore size ( $V_c$ ), and another morphology with gradual cell size ( $V_g$ ), following a Voronoi tessellation technique presented in section 2. Furthermore, and in order to have a broader comparison with the new Voronoi morphology, three lattices evaluated during the first campaign were tested as well; the best unstructured lattice ( $F_s$ ), the best periodical cellular lattice ( $T_s$ ) and the reference honeycomb absorber (H).

The main difference between the gradual Voronoi lattice ( $V_g$ ) and the gradual lattices tested during the first campaign, is that the Voronoi lattice ( $V_g$ ) was manufactured in a single piece as shown in Fig. 5 (a), while the previous lattices consisted of the assembly of each desired pore size lattice to obtain a compound structure, Fig. 5 (b)–(c). For the increasing cell size distribution (s-m-b), the fluid flows from small-s to big-b pores, while in the decreasing (b-m-s) cell size the fluid flows from big-b to small-s pores, each piece having a constant pore size.

To determine the feasibility of these innovative cell structures, a thermodynamic analysis was conducted under concentrated solar radiation conditions.





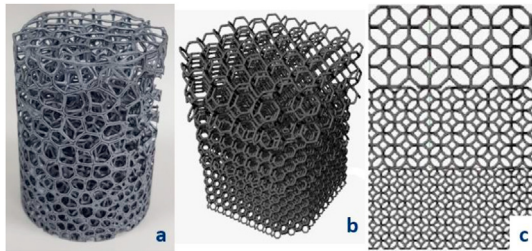
**Fig. 4.** SiSiC volumetric absorbers with different morphologies: a) Diamond (D), b) random Foam (F), c) Tetrakaidecahedron (T), d) Rotated Cube (RC), e) Cube Shifted (CS), f) Voronoi with constant cell ( $V_c$ ), g) Voronoi with gradual cell ( $V_g$ ), h) Honeycomb (H).

**Table 1**

Absorbers tested during each test campaign.

| Test campaign | Pore distribution | D                 | T         | RC                  | CS   | F         | H | $V_c$ | $V_g$ |
|---------------|-------------------|-------------------|-----------|---------------------|------|-----------|---|-------|-------|
| First         | Constant          | $D_s$             | $T_s$     | $RC_s$              | CS   | $F_s$     | H | n.a.  | n.a.  |
|               | Gradual           | $D_{bms}/D_{smb}$ | $T_{bms}$ | $RC_{bms}/RC_{smb}$ | n.a. | $F_{bms}$ | — | n.a.  | n.a.  |
| Second        | Constant          | —                 | $T_s$     | —                   | —    | $F_s$     | H | $V_c$ | n.a.  |
|               | Gradual           | —                 | —         | —                   | —    | —         | — | n.a.  | $V_g$ |

n.a. — not available.



**Fig. 5.** Absorbers: a) 3D- $V_g$ , b) 3D- $T_{bms}$ , c) 2D- $T_{bms}$ .

### 2.3. Properties of the morphologies

The thermal and hydrodynamic performance of volumetric absorbers is highly influenced by geometrical and optical parameters; the specific surface area ( $\alpha_v$ ), the porosity ( $\emptyset$ ) and, the extinction coefficient ( $\beta$ ). The three parameters are related to the morphological structure of the absorber, which generally has a large specific surface area and a highly tortuous flow path to enhance the heat transfer between the solid matrix and the circulating fluid. The geometrical parameters, in determining the specific surface area drive the convective heat transfer mechanisms [52], while the optical parameters, in determining the extinction coefficient, drive the penetration depth of the incident radiation in the absorber [53].

Furthermore, previous authors [38,54] have stated that the dominant factors in absorber thermal performance are the heat transfer coefficient and the radiation penetration depth (inverse of the extinction coefficient), and both are functions of the absorbers geometrical parameters. However, despite the interest in improving both parameters, they diverge, such that increasing the porosity and cell size in random foams has resulted in decreasing the heat transfer coefficient and increasing the penetration depth ( $1/\beta$ ).

#### 2.3.1. Geometrical parameters

An accurate estimation of the geometrical porous parameters undoubtedly helps to understand the convective heat transfer mechanisms inside porous structures. This section presents the volumetric porosity, and the specific surface area.

**2.3.1.1. Porosity.** The volumetric porosity,  $\emptyset$ , of a porous structure is defined as the ratio of the void volume to the total volume of a cell.

$$\emptyset = 1 - \frac{V_s}{V_{tot}} \quad (1)$$

**2.3.1.2. Specific surface area.** The specific surface area,  $\alpha_v$ , of a porous matrix is defined as the ratio of the total surface area to the total volume.

$$\alpha_v = \frac{S_s}{V_{tot}} \quad (2)$$

#### 2.3.2. Optical parameters

Porous materials, used as volumetric absorbers, work at very high operational temperatures, so heat transfer by radiation has an important impact in absorber performance, especially on the frontal surface of the absorber by means of the radiative heat losses. The main optical parameters used in porous media, especially for modeling the radiative transfer equation, are the absorption coefficient, the scattering coefficient and the extinction coefficient.

The optical or radiative properties associated with porous materials have been extensively studied in the literature for both ceramic [55–59] and metallic materials [60–62] by means of experiment, the Monte Carlo ray-tracing technique and geometrical optics.

In this research, and as a first approximation, the optical parameters were inferred using the geometrical optic approximation [7], which used the pore diameter, porosity and material absorptivity as its major empirical parameters.

**2.3.2.1. Absorption coefficient.** The absorption coefficient is the property of the medium that describes the amount of absorption of thermal radiation per unit path length within the medium. It can be interpreted as the inverse of the mean free path that a photon will travel before being absorbed. The absorption coefficient is:

$$a = 1.5 \cdot \alpha \cdot \frac{(1 - \varnothing)}{d} \quad (3)$$

Where  $a$  is the absorption coefficient,  $\alpha$  is the material absorptivity,  $\varnothing$  is the porosity, and  $d$  is the cell size.

**2.3.2.2. Scattering coefficient.** The scattering coefficient is the property of a medium that describes the amount of scattering of thermal radiation per unit path length for propagation in the medium. It can be interpreted as the inverse of the mean free path that a photon will travel before undergoing scattering. The scattering coefficient is:

$$\sigma_s = 1.5 \cdot (2 - \alpha) \cdot \frac{(1 - \varnothing)}{d} \quad (4)$$

Where  $\sigma_s$  is the scattering coefficient,  $\alpha$  is the material absorptivity,  $\varnothing$  is the porosity, and  $d$  is the cell size.

**2.3.2.3. Extinction coefficient.** The extinction coefficient is the sum of absorption coefficient and scattering coefficient and measures the capacity of a porous medium to absorb and scatter the radiation. The extinction coefficient is:

$$\beta = a + \sigma_s = 3 \cdot \frac{(1 - \varnothing)}{d} \quad (5)$$

Where  $\beta$  is the extinction coefficient,  $\varnothing$  is the porosity, and  $d$  is the cell size.

The main parameters depend on the geometrical parameters and the optical absorptivity of the porous material. The geometrical parameters of each single unit cell of the different lattices were analysed by means of CAD files ( $\varnothing$ ,  $a_v$ ), experimental measures of the lattices ( $d_{st}$ ,  $d_c$ ) and geometrical optics ( $\beta$ ), and the summary

results are presented in Table 2.

All the single and small cell size morphologies have similar porosity and extinction coefficients, except the cube shifted, which exceeds 90% porosity and has the lowest extinction coefficient. Concerning the specific surface area, random foam has the largest value of all the samples tested, while tetrakaidecahedron ( $T_s$ ) and rotated cube ( $RC_s$ ) are the lattices with the largest values, followed by  $V_c$  and  $CS$ . The diamond ( $D$ ) lattice is the one with the lowest specific surface area.

The gradual configurations are based on varying the cell size in an increasing and decreasing pattern in the direction of the fluid flow with the associated geometrical and optical properties presented in Table 2.

### 3. Experimental setup

#### 3.1. Facility description

A 1 kW thermal power test bench is used for the experimental evaluation of the different innovative silicon carbide morphologies. The main associated advantage to this facility is the possibility of carrying out quick and effective screening on new absorbers before higher prototypes are tested at larger facilities, i.e. solar furnace, parabolic dishes or power tower systems. CIEMAT-PSA has broader knowledge in the experimental testing and evaluation of different sized volumetric absorbers, from laboratory scale (~1 kW thermal), medium scale (~250 kW thermal) to pre-commercial prototypes (~3000 kW thermal).

The open volumetric absorber facility presented in Fig. 6 contains four sub-components for a comprehensive thermal evaluation of the absorbers, which are:

- A 4 kW electrical power solar simulator made up of a xenon lamp and a parabolic concentrator. Further details about the xenon lamp have been described by Gallo [63].
- A cylindrical receiver sub-system where the volumetric absorbers of 40, 50 or 60 mm diameter are placed. It is equipped with 24 K-type thermocouples distributed in six sections, 1 T-type thermocouple, 2 PT100 surface sensors and an infrared camera.
- A helical air-water heat exchanger sub-system equipped with 4 PT100 sensors, 2 PT100 surface sensors, 1 T-type thermocouple, a water mass flow-rate measurement, and a pump.

**Table 2**  
Geometrical parameters of the tested absorbers.

| Lattice                                     | $D_s$ | $T_s$ | $RC_s$ | $CS$ | $F_s$ | $V_c$ | $D_{bms}$                 | $D_{smb}$ | $RC_{bms}$                | $RC_{smb}$ | $T_{bms}$            | $F_{bms}$            | $V_g$                      |
|---------------------------------------------|-------|-------|--------|------|-------|-------|---------------------------|-----------|---------------------------|------------|----------------------|----------------------|----------------------------|
| $\varnothing$ , %                           | 87.7  | 87.8  | 87.6   | 93.2 | 86.7  | 86.0  | ↓<br>97.6<br>96.5<br>87.7 | ↑         | ↓<br>94.4<br>95.1<br>87.6 | ↑          | 95.0<br>89.6<br>87.8 | 89.6<br>89.4<br>86.7 | 91.5<br>84.0 <sup>a</sup>  |
| $\alpha_v$ , m <sup>2</sup> /m <sup>3</sup> | 133   | 754   | 750    | 400  | 1150  | 470   | ↓<br>83<br>138<br>133     | ↑         | ↓<br>267<br>377<br>750    | ↑          | 204<br>457<br>754    | 480<br>740<br>1150   | 320<br>800 <sup>a</sup>    |
| $d_{st}$ , mm                               | 1.21  | 0.99  | 0.88   | 1.62 | 0.70  | 1.12  | ↓<br>1.70<br>1.25<br>1.21 | ↑         | ↓<br>1.37<br>1.03<br>0.88 | ↑          | 1.45<br>1.22<br>0.99 | 0.76<br>0.70<br>0.70 | 1.27<br>0.84 <sup>a</sup>  |
| $d_c$ , mm                                  | 3     | 3     | 3      | 8    | 3     | 5     | ↓<br>10<br>6.5<br>3       | ↑         | ↓<br>10<br>6.5<br>3       | ↑          | 10<br>6.5<br>3       | 10<br>6.5<br>3       | 11<br>3.5 <sup>a</sup>     |
| $\beta$ , 1/m                               | 123   | 122   | 124    | 25.5 | 133   | 84    | ↓<br>7.2<br>16.2<br>123   | ↑         | ↓<br>16.8<br>22.6<br>124  | ↑          | 15<br>48<br>122      | 31.2<br>48.9<br>133  | 23.2<br>137.1 <sup>a</sup> |

<sup>a</sup> Linear variation.

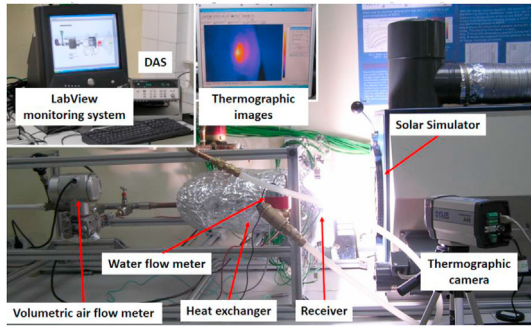


Fig. 6. Open volumetric receiver equipment.

- d) An extraction sub-system with 1 PT100 sensor, an air mass flow-rate meter, and a blower.

The experimental procedure is as follows; firstly, a Gardon radiometer, with an error of less than 3%, is used to measure the concentrated flux over the sample, secondly, the 60 mm diameter absorber is placed in the receiver system and the data acquisition system and infrared camera are connected, thirdly, the solar simulator, the extraction sub-system, which is at the higher air mass flow rate, are turned on. Finally, the air-water heat exchanger is connected and linked to the previous components. Further information of the facility could be found in previous works of Avila [37,64].

The radiation is concentrated by an elliptical reflector and then strikes the absorber sample. Simultaneously, the blower forces the ambient air to flow through the volumetric absorber as it heats up. The air is the energy carrying fluid which circulates counter current in the water heat exchanger in order to reduce the temperature of the heated air, protect the blower and get an indirect measure of the receiver's efficiency. Although the air temperature is measured continuously with the 24 thermocouples installed in the receiver, as a result of the inhomogeneous concentrated radiation and the difficulties associated with the measurement of the air mass flow rate at high temperatures, a heat exchanger system has to be used in order to determine the total power gained by the air in the receiver. The water mass flow rate was adjusted in order to get a significant temperature difference between upstream and downstream, measured with two type-A PT-100 sensors.

The test of each sample took from 4 to 5 h. During this time the irradiation was kept constant. To achieve various air outlet temperatures, the air mass-flow was varied three times. At the start-up, the highest value (10 m<sup>3</sup>/h) was set to avoid thermal shock. After each step the mass flow was kept constant for 90–120 min to reach thermal equilibrium. The reason for this long period is that, despite the air quickly achieving a steady state, the water in the heat exchanger needed more time to reach a steady condition. Once that was achieved, the temperatures and the radiation data were taken to characterise this steady state period. After 3 steps the lowest mass flow (6 m<sup>3</sup>/h) was set to the corresponding “hottest” operating point. Then, each sample was tested at least two more times to demonstrate the general repeatability of the efficiency measurement.

### 3.2. Open volumetric absorber performance

The main goal of this experimental research is to get the thermodynamic performance of each absorber and the comparison

among them. Then the best choice will be selected for further investigation at PSA facilities. For the comparison of the different absorbers three variables are being used: 1) absorber efficiency; 2) absorber air outlet temperature and; 3) the maximum solid temperature at the absorber aperture. Variables 1 and 2 are obtained by a thermal evaluation of the experimental results presented in the next sections, while the third variable is obtained by direct measurement with a thermographic camera. Finally, in order to analyse the real improvement of the new structures, a comparison with the state of the art is required. Thus, the well-known SOLAIR absorber (H), which consists of a silicon carbide honeycomb absorber [65] with square channel aperture, is being tested at the same operational conditions as the previous lattices and is considered as the reference absorber.

#### 3.2.1. Absorber efficiency

Absorber efficiency,  $\eta_{abs}$ , is defined as the ratio between the power transferred by the absorber to the air and the incident power in the frontal surface of the receiver [66]:

$$\eta_{abs} = \frac{\dot{m} \cdot \int_{T_{amb}}^{\bar{T}_{a-abs-out}} c_{p_a} \cdot dT_a}{\bar{P}} \quad (6)$$

Thermal efficiency is a function that depends on the morphological design of the absorber, the experimental conditions (orientation, distance to the focal plane, etc.) and the operational working conditions (output temperature regimen, input power flux, etc.). At lab scale, most of the working and operational conditions are controlled, so the methodology of the test carried out consisted of the variation of the volumetric mass flow rate,  $\dot{m}$ , under similar incident flux.

The incident flux,  $\varphi$ , was measured by a calibrated water-cooled calorimeter, and the evaluation of the average incident flux,  $\bar{\varphi}$ , and the average incident power,  $\bar{P}$ , were calculated with equations (7) and (8).

$$\bar{\varphi} = \int_{-y}^y \int_{-x}^x \varphi(x, y) \cdot dx \cdot dy \quad (7)$$

$$\bar{P} = \bar{\varphi} \cdot A_{abs} \quad (8)$$

where  $\bar{\varphi}$  (W/m<sup>2</sup>) is the average incident flux measured in a constant plane,  $\bar{P}$  (W) is the average incident power and,  $A_{abs}$  (m<sup>2</sup>) is the frontal surface.

#### 3.2.2. Absorber air outlet temperature

Although 24 thermocouples were installed in the test-bed, it is not possible to know exactly the average air outlet temperature due to the inhomogeneous incident flux map over the absorbers. So it is necessary to estimate it using the following methodology:

- Calculation of the power gained by the water in the heat exchanger:  $P_{w-HE} \cong P_{a-HE}$
- Calculation of the air outlet temperature from the receiver:  $\bar{T}_{a-rec-out}$
- Calculation of the air power at the absorber outlet:  $P_{a-abs}$
- Calculation of the air outlet temperature from the absorber:  $\bar{T}_{a-abs-out}$

**3.2.2.1. Power gained by the water in the heat exchanger.** The heat exchanger system usually considers that the air sensible heat matches the water sensible heat at steady conditions as presented in Eq. (9):

$$P_{a-HE} \cong P_{w-HE} = \dot{m}_w \cdot \int_{T_{w-out-HE}}^{T_{w-in-HE}} c_{p_w}(T_w) \cdot dT_w \quad (9)$$

where  $P_{a-HE}$  (W) is the power gained by the air in the receiver and transmitted into the heat exchanger,  $P_{w-HE}$  (W) is the power gained by the water in the heat exchanger,  $\dot{m}_w$  (kg/s) is the water mass flow rate,  $c_{p_w}$  (J/(kg · K)) is the heat capacity of the water at constant pressure,  $T_{w-in-HE}$  (K) is the heat exchanger inlet water temperature and,  $T_{w-out-HE}$  (K) is the heat exchanger outlet water temperature.

**3.2.2.2. Calculation of the air outlet temperature from the receiver.** Once the power gained by the air in the receiver is computed with expression (9), the air temperature at the receiver outlet or the heat exchanger inlet ( $\bar{T}_{a-rec-out}$ ), is obtained using iteration in the following energy balance.

$$P_{a-HE} \cong P_{w-HE} = \dot{m}_w \cdot \int_{T_{w-out-HE}}^{T_{w-in-HE}} c_{p_w}(T_w) \cdot dT_w = \dot{m}_a \cdot \int_{T_{amb}}^{\bar{T}_{a-rec-out}} c_{p_a}(T) \cdot dT(T_{amb}; \bar{T}_{a-rec-out}) \quad (10)$$

For the application of this energy balance, the temperature of the air at the exit of the heat exchanger is used, which is nearly the ambient temperature.

**3.2.2.3. Power gained by the air in the absorber.** The power gained by the air in the absorber is the sum of the power of the air delivered in the heat exchanger and the thermal losses throughout the receiver.

$$P_{a-abs} = P_{a-rec} + TL_{rec} \quad (11)$$

Where  $P_{a-abs}$  (W) is the power gained by the air in the absorber and  $TL_{rec}$  (W) is the thermal loss in the receiver.

Two PT100 surface sensors were installed in the receiver outer surface (the first in the upper-front surface and the second in lower-back surface) and measure the surface temperature in the receiver system. The average temperature were used for the estimation of the thermal losses by convection ( $TL_{conv,rec}$ ) and radiation ( $TL_{rad,rec}$ ) following next expression:

$$TL_{rec} = TL_{conv,rec} + TL_{rad,rec} \quad (12)$$

**3.2.2.3.1. Thermal losses by convection.** The thermal loss by convection through the receiver is evaluated with the following expression:

$$TL_{conv,rec} = \pi \cdot d_{bk-rec-ext} \cdot L_{rec} \cdot h_{ext} \cdot (\bar{T}_{rec-ext} - T_{amb}) \quad (13)$$

As the tests are performed at lab scale, there is no wind and the convection heat transfer from the receiver to the environment is due to natural convection. For this case, the correlation developed by Churchill is used to estimate the Nusselt number [67].

$$Nu_{bk} = \left\{ 0.60 + \frac{0.387 \cdot Ra_{bk}^{1/6}}{\left[ 1 + (0.559/Pr_{bk})^{9/16} \right]^{8/27}} \right\}^2 \quad (14)$$

Where  $Nu_{bk}$  is the dimensionless Nusselt number,  $Ra_{bk}$  is the dimensionless Rayleigh number for the air based on the blanket outer diameter ( $d_{bk-rec-ext}$ ),  $Pr_{bk}$  is the dimensionless Prandtl number. This correlation is valid for  $10^{-5} < Ra \leq 10^{12}$  and  $0 < Pr < \infty$ .

**3.2.2.3.2. Thermal losses by radiation.** The thermal loss by radiation through the receiver is caused by a temperature difference between the blanket and the environment. To approximate this, the envelope is assumed to be a small convex grey object that emits to the environment. The net radiation transfer becomes [67]:

$$TL_{rad,rec} = \pi \cdot d_{bk-rec-ext} \cdot L_{rec} \cdot \epsilon_{bk} \cdot \sigma \cdot (\bar{T}_{rec-ext}^4 - T_{amb}^4) \quad (15)$$

Where  $\epsilon_{bk}$  is the emissivity of the blanket,  $\sigma$  (W/m<sup>2</sup> · K<sup>4</sup>) is the Stefan-Boltzmann constant number.

**3.2.2.4. Calculation of the air outlet temperature from the absorber**

Once the power gained by the air in the absorber is known with equation (11), the mean air outlet temperature from the absorber ( $\bar{T}_{a-abs-out}$ ) can be calculated using iteration in the following energy balance:

$$P_{a-abs} = \dot{m}_a \cdot \int_{T_{amb}}^{\bar{T}_{a-abs-out}} c_{p_a}(T) \cdot dT(T_{amb}; \bar{T}_{a-abs-out}) \quad (16)$$

**3.2.3. Maximum solid temperature at the absorber inlet**

The measurement of the solid absorber temperature at the irradiated surface is a complex measurement that is affected by different parameters, such as the emissivity of the sample, the concentrated solar flux, the surrounding temperature, the camera filters, the physical separation between the camera and the absorber, humidity, etc. During the tests, a FLIR thermographic camera model A40M, with a temperature ranging from 300 to 1500 °C was used to monitor the absorber's frontal temperature.

Due to the inhomogeneous concentrated solar flux, presented in Fig. 7, the solid temperature variation is very intense, especially at the frontal surface. Thus, in this work, the maximum solid temperature at the absorber inlet is registered and considered as a magnitude for comparison between different absorbers, to analyse the influence of the different characteristic parameters of the absorbers and the change in the temperature. Fig. 8 shows some experimental measurements of the inlet absorber temperature for diamond lattice, random foam, and Voronoi morphologies.

Concerning the concentrated solar flux, it was observed that during the second test campaign the average concentrated “solar” power of the solar simulator was slightly lower, 788 W, compared to the average value during the first tests of 819 W.

## 4. Results and discussion

The 3D printed and thermally treated configurations presented in Fig. 4 have been tested in the open volumetric receiver facility in



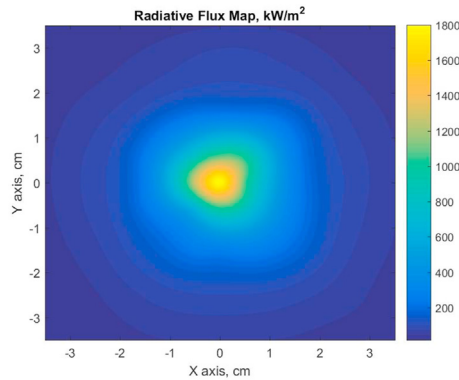


Fig. 7. Representative experimental **concentrated flux profile**.

two different campaigns, and three main parameters have been used to evaluate and compare the different samples: i) thermal absorber efficiency (Eq. (6)), ii) absorber mean air outlet temperature (Eq. (16)) and iii) the maximum solid inlet temperature. Table 1 shows the absorbers tested during the two test campaigns. All the morphologies were tested at least three times at each volumetric flow rate, and the average value of each magnitude is adopted for comparison.

#### 4.1. First test campaign

Ceramic honeycombs and unstructured foams have been the most widely adopted choices for high-temperature volumetric solar absorbers. While these two types of porous structure are commercially available and offer some flexibility in pore scale geometry, it was not clear if they are the best choice in terms of thermal performance [68]. The current research studied experimentally, for the first time, four innovative periodical cellular structures as volumetric receivers: Diamond cell (D), Tetraikahedron cell (T), Rotated Cube cell (RC), and Cube Shifted cell (CS). Moreover, the two most popular choices adopted so far were tested as well, a piece of the SOLAIR receiver with a square channel-shape honeycomb (H) configuration and a ceramic random foam (F).

Besides testing each of the samples with the small pore size as the most promising cell size, three (D-T-RC) of the four new morphologies were tested with a gradual, increasing and/or decreasing cell size configuration. The fourth sample (CS) was manufactured in only one cell size, which obviated any other possible configuration.

##### 4.1.1. Gradual cell size configurations

Firstly, the gradual cell size configurations were tested in order to reach preliminary conclusions. Several previous works have analysed the influence of different gradual configurations. Roldan [69] numerically studied a local thermal equilibrium model (LTE), which only takes into account the fluid temperature, the influence of gradually increasing and decreasing porosity configurations according to depth over a honeycomb absorber, concluding that the decreasing-porosity variation performs better than the increasing-porosity variation absorber. Chen [70] studied the influence of gradually increasing and decreasing porosity and cell size configurations in a double-layer ceramic foam with the local thermal non-equilibrium (LTNE) model. Chen reached similar conclusions to Roldan regarding the gradual porosity configurations. However, his analysis of the gradual cell size configurations showed that the increasing (from 1 to 3 mm) cell size design improved performance compared to the decreasing pattern. Avila [37,64] studied experimentally the effect of decreasing pore size configurations over wire mesh volumetric absorbers, showing that this design is able to improve the state of the art of volumetric absorbers.

Fig. 9 and Fig. 10 show the thermal evaluation of the single and small cell, the increasing and the decreasing cell size pattern for the diamond (D) and rotated cube (RC) morphologies, respectively. Figs. 9 (a) and Fig. 10 (a) depict the absorber thermal efficiency variation using the air temperature at the absorber outlet. These figures (a) present scatter values of the different tests repetitions (black symbols) and the average value of the different repetitions (blue symbols with blue line). Figs. 9 (b) and Fig. 10 (b) presents the variation in the maximum solid temperature at the frontal surface of the absorber with the air temperature at the absorber outlet. The average values of both magnitudes are presented with blue symbols together with the deviation for each variable (the red bar refers to the air outlet temperature while the black bar refers to the maximum solid temperature). Moreover, lattices with single cell sizes ( $D_s$  and  $RC_s$ ) have an overall thickness of ~26 mm, while gradual configurations lattices, made after assembling three different pieces in gradual cell size pattern and in both directions (b-m-s and vice versa), have a total thickness of ~80 mm.

The results show a clear tendency for both unit cells (D and RC) where the single and small (s) cell size morphology have the best thermal efficiency together with the higher maximum solid temperature at the frontal surface. Among them,  $D_s$  and  $RC_s$ , the rotated cube have a slightly better thermal efficiency than the diamond lattice, with an improvement ranging from 0.4-to-1.0%. Concerning the maximum solid temperature, it is slightly higher for  $RC_s$  lattice

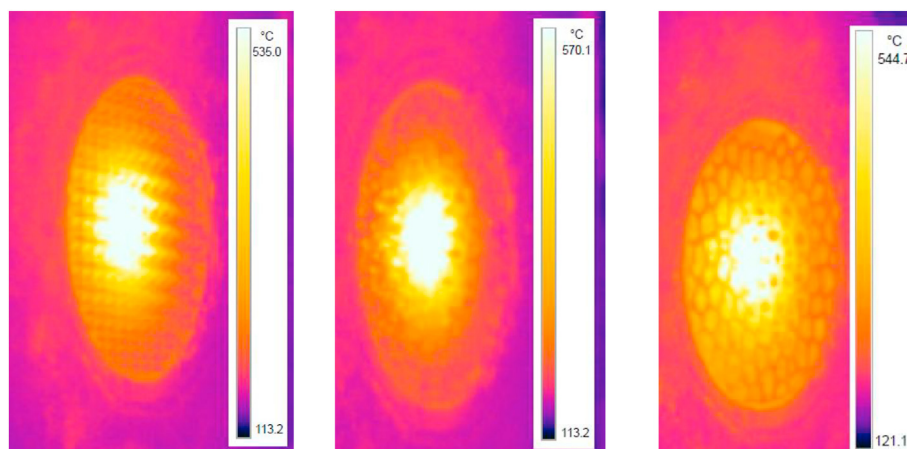
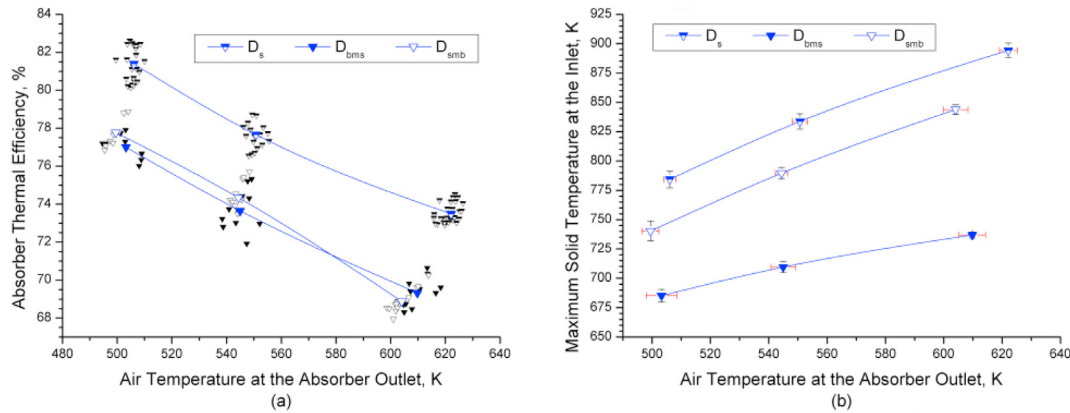
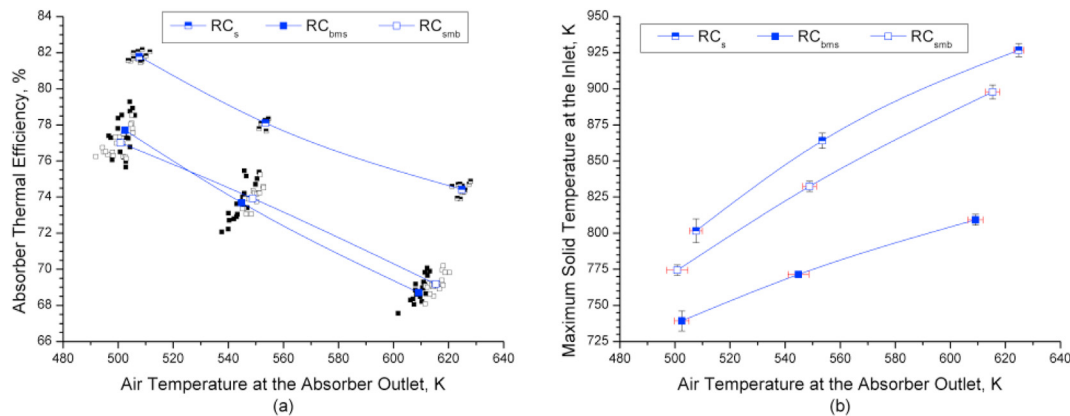


Fig. 8. Experimental measurements of the frontal solid temperature for diamond cell (left), random foam (center), and Voronoi (right).





**Fig. 9.** Diamond (D) lattice results for single and triple –increasing and decreasing– cell size: (a) absorber thermal efficiency and air outlet temperature, (b) maximum solid inlet temperature and air outlet temperature.



**Fig. 10.** Rotated cube (RC) lattice results for single and triple –increasing and decreasing– cell size: (a) absorber thermal efficiency and air outlet temperature, (b) maximum solid inlet temperature and air outlet temperature.

with a temperature increase between 17 and 33 K.

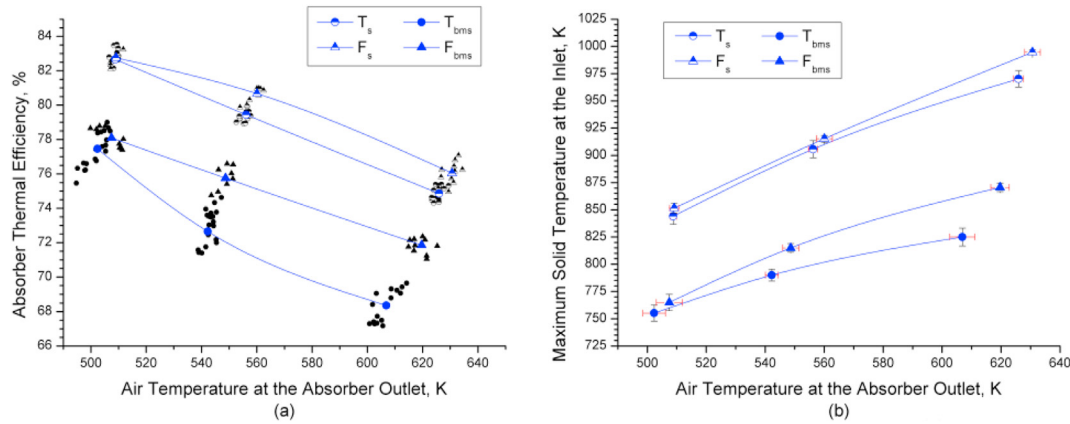
In general, the thermal performance of volumetric receivers is mainly affected by two heat transfer mechanisms, the radiation heat transfer according to the extinction coefficient and the convective heat transfer related to the specific surface area. Both parameters, the extinction coefficient and the specific surface area, depend on the absorber morphology and have opposite trends. Here both lattices ( $D_s$  and  $RC_s$ ) have similar porosities, specific surface areas and cell diameters, while the specific surface area is ~5 times higher for the rotated cube unit (Table 2), showing that among both morphological lattices the specific surface area gives better thermal results regarding thermal efficiency, despite having a slightly higher solid temperature.

Furthermore, increasing or decreasing cell sizes clearly worsens the thermal performance of the single and small-cell-size lattices, both patterns having similar thermal efficiencies as seen in Figs. 9 (a) and Fig. 10 (a) and, lower maximum solid temperatures than the single lattices (Figs. 9 (b) and Fig. 10 (b)). The decreasing cell pattern (b-m-s) has the lowest extinction coefficient and specific surface area (Table 2) at the absorber inlet favouring a higher solar penetration depth that reduces the maximum solid temperature at the frontal surface, penalizing the convective mechanism. The increasing cell pattern (s-m-b) shows the other tendency, having the maximum possible specific surface area and extinction coefficient, favouring the convective heat transfer while penalizing the penetration depth due to a larger path. Hence, with the available cell size lattices, each pattern follows a trend that clearly favours

one of the two main heat transfer mechanisms while penalizing the other to finally achieve a similar thermal performance in terms of thermal efficiency, having lower solid temperature for the decreasing than for the increasing pattern. These results demonstrate that having a lower solid temperature at the frontal surface does not necessarily mean having a higher thermal efficiency, as a higher penetration depth does not indicate a higher heat transfer coefficient. Finally, it is worth mentioning that the gradual cell size configurations have a solar path three times larger (~80 mm) than the single cell size absorber (~26 mm), which also influences the thermal performance of the lattice, as the accumulated heat throughout the absorber is increased without being properly transferred to the heat transfer fluid.

Fig. 11 shows the performance of the single and small cell size and the decreasing (b-m-s) cell size patterns for the lattice made with the tetrakaidecahedron (T) unit and the unstructured random foam (F). Fig. 11 (a) presents the absorber thermal efficiency variation with the air temperature at the absorber outlet. The scatter values reflect the test repetitions and the line-symbol combination depicts the average values for the different tests. Fig. 11 (b) presents the maximum solid temperature at the absorber inlet against the average air temperature leaving the absorber. Both magnitudes show the deviation of each variable.

Comparing the  $T_s$  and  $R_s$  absorbers and checking their properties in Table 2, both can be seen to have similar porosities, cell sizes and extinction coefficients, while the specific surface area is 1.5 higher for the random foam. Moreover, the thermal efficiency of the



**Fig. 11.** Tetra- and random foam (F) results for single and triple –increasing– cell size: (a) absorber thermal efficiency and air outlet temperature, (b) maximum solid inlet temperature and air outlet temperature.

random foam is higher than the tetra lattice, ranging from ~0.1 to 1.2%, while the maximum solid temperatures at the matrix inlet are quite similar for both. Thus, once the samples have similar geometrical and optical properties, the specific surface area determines the absorber behaviour.

Comparing each configuration with its gradual configuration counterpart, the gradual decreasing cell size tetra- and random foam absorbers can be seen to have worse overall performance than the single and small cell size configurations. At the same time, the decreasing cell size arrangements have lower maximum solid temperatures compared to the single cell size absorbers, with temperature reductions ranging from 90 to 145 K for tetra- and random foam lattice and from 85 to 125 K for random foam.

The parameters presented in Table 2 show that the lower extinction coefficient within the first absorber piece, for both absorbers, favours a reduction in the maximum frontal solid temperature compared to the single cell absorbers. However, the specific surface area of this piece and the larger flow path obviate effective heat transfer, such that thermal efficiency is clearly reduced.

#### 4.1.2. Single pore size configurations

Once gradual cell sizes with increasing and decreasing patterns were verified to be unable to improve the single and small size lattices, a comparison of the innovative structured single unit cells (T-D–CS–RC), together with conventional or baseline absorbers (periodical silicon carbide honeycomb-H and unstructured random foams-F), was undertaken whose results are presented in Fig. 12. It is worth noting, that both baseline absorbers (H and  $F_s$ ) are the ones with the best thermal performance, showing why most of the recent works have been focused on them.

A visual inspection of all the lattices manufactured for this study (T-D–CS–RC), including the random foam (F), revealed a brighter colour than the SiC honeycomb, which negatively affects its thermal performance because of the expected higher reflectivity that increases the optical losses.

Among the new lattices (T-D–CS–RC), the Cube Shifted configuration shows the poorest thermal performance together with the lowest maximum solid temperature. Moreover, as shown in Table 2, it is the lattice with the lowest extinction coefficient and the second lowest specific surface area. The Diamond lattice has the second worst performance together with the second lowest maximum solid temperature at the absorber inlet. This lattice has the lowest specific surface area of all the single and small cell size specimens while having an extinction coefficient similar to the

remaining lattices. Thus, a lattice having the lowest values of any on the main heat transfer parameters, extinction coefficient or specific surface area, will have the poorest thermal performance.

On the other hand, the best absorber of the configurations manufactured for this research and in terms of thermal efficiency, is the random foam (F), which has an unstructured intricate morphological arrangement. This absorber has an extinction coefficient around  $130 \text{ m}^{-1}$ , similar to the other lattices, while having the largest specific surface area ( $\sim 1100 \text{ m}^2/\text{m}^3$ ). However, it is the absorber with the highest maximum solid temperature demonstrating again that having a higher solid temperature at the absorber inlet does not mean having a lower thermal performance.

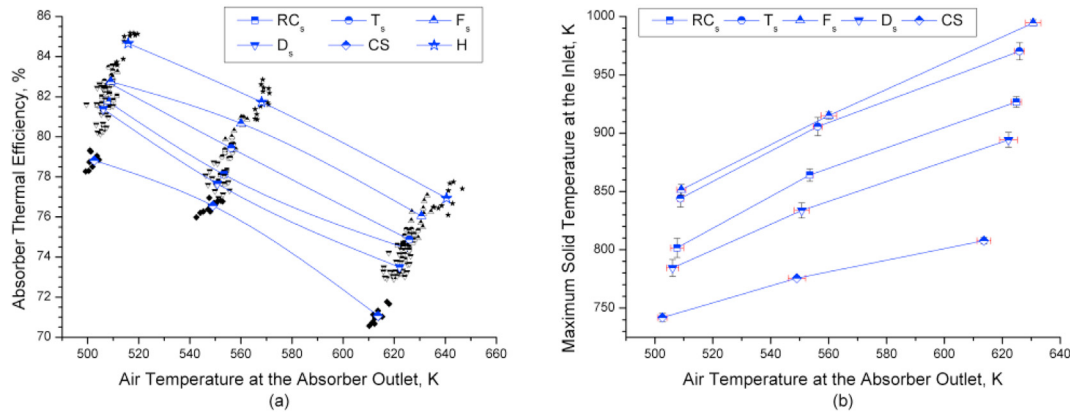
Finally, the remaining configurations are the tetra- and random foam (T) and the rotated cube (RC), and both have very similar parameters. Between them, the tetra lattice has a better thermal efficiency ranging from 0.4- to 1.3%, and a higher maximum frontal solid temperature as well. In this case, the morphological configuration of the lattice which somehow tends to be chaotic, similarly to unstructured foams, and which tends to play a significant role in its thermal performance compared to the rotated cube lattice.

#### 4.2. Second test campaign

In parallel with the first test campaign and knowing the preliminary thermal results of the new lattices, a new morphology, called Voronoi (V) morphology [71,72], was developed and suggested as a potential absorber candidate. This morphology results from a manufacturing process that allows slight variations in the cell sizes/geometrical parameters to be distributed non-uniformly within the lattice, unlike the previous lattices, which require assembling different pieces to obtain a variation in cell size. The Voronoi morphology cannot be classified as a lattice or a structured absorber. It is more like a random foam manufactured with an improved design technique called Voronoi tessellation.

In order to compare the thermal results of the new morphology, V, the main samples tested in the first campaign were tested again under the same working conditions with the aim of obtaining a broader range of comparison: the baseline ceramic absorber with honeycomb morphology (H) which was the absorber with the highest thermal performance, the conventional unstructured foam ( $F_s$ ), and the best structured morphology,  $T_s$ . So, five configurations were tested, as presented in Table 1, during the second campaign:  $T_s$ ,  $F_s$ , H, V, and  $V_g$ .

Fig. 13 (a) shows the absorber thermal efficiency variation and Fig. 13 (b) shows the variation of the maximum solid temperature,



**Fig. 12.** Tetrakaidecahedron (T), Reticulated Cube (RC), Diamond (D), Cube Shifted (CS), random foam (F) and Honeycomb (H) absorber results for single and small cell size: (a) absorber thermal efficiency and air outlet temperature, (b) maximum solid inlet temperature and air outlet temperature.

both as a function of the average temperature of the air at the absorber outlet. It is noteworthy that for the first time in this research, a new morphology with two different patterns,  $V_c$  and  $V_g$ , is able to improve on the thermal performance of baseline absorbers: the random foam with small cell size ( $F_s$ ) and the structured honeycomb absorber (H). It is also worth noting that the two absorbers with the highest thermal efficiency are the ones with the lowest maximum solid temperature at the absorber inlet, in contrast to the previous absorbers.

The quantitative results of thermal efficiency presented in Fig. 13 (a), shows that the **Voronoi morphology with gradual cell configuration ( $V_g$ ) compared to the honeycomb structure (H), have a higher air outlet temperature of about  $+10/+13$  °C, and a higher thermal absorber efficiency of  $+1.9/+3.7\%$ , and compared to the random foam ( $F_s$ ), have a higher air outlet temperature of about  $+21/+25$  °C, and a higher thermal absorber efficiency of  $+4.1/+5.5\%$ .**

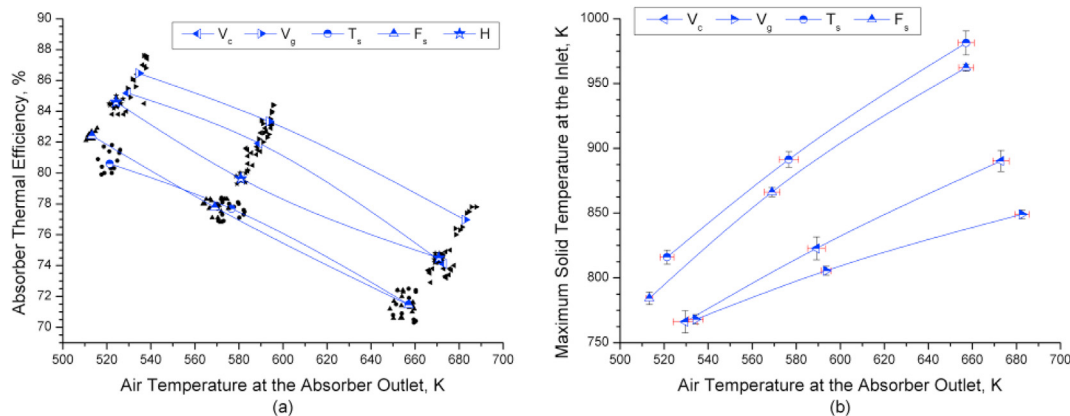
The configuration that performs the best,  $V_g$ , follows a linear variation within the morphology of all the geometrical and optical parameters, starting at 91.5% porosity, an 11 mm cell size, a  $320\text{ m}^{-1}$  specific surface area and a  $23\text{ m}^{-1}$  extinction coefficient (inlet surface) and ending at 84% porosity, a 3.5 mm cell size, an  $800\text{ m}^{-1}$  specific surface area and a  $137\text{ m}^{-1}$  extinction coefficient (outlet surface). The inlet parameters of  $V_g$  are quite similar to those of the CS morphology (Table 2), which was the worst thermal absorber previously, and the outlet parameters are quite similar to the  $T_s$  and  $RC_s$  morphologies, which previously presented an intermediate

thermal value efficiency (Fig. 12). The second best configuration,  $V_c$ , has constant parameters having 86% porosity, a 5 mm cell size, a  $470\text{ m}^{-1}$  specific surface area and an  $84\text{ m}^{-1}$  extinction coefficient.  $V_c$  has a porosity similar to all the other morphologies, a cell diameter higher than the usual 3 mm and intermediate values for the specific surface area and the extinction coefficient.

Thus, in the case  $V_g$ , it is observed that having a linear variation of the parameters obviously favours an optimal penetration depth for the solar radiation inside the morphology, while the heat transfer to the air is improved. However, this absorber does not excel in any of the parameters, but rather in balancing all the parameters throughout the absorber. In the  $V_c$  case, similarly to the  $V_g$  case, an intermediate value of the specific surface area and the extinction coefficient appears to be a good balance for improving on the performance of the state of the art in volumetric absorbers. Furthermore, both configurations with Voronoi morphology seem to provide enhanced performance compared to previous lattices because of its intricate and almost random morphology improved with the upgraded Voronoi tessellation technique implemented.

Finally, to be best of the authors' knowledge, this is the first time that a linear cell size variation within an absorber has been manufactured and tested experimentally.

Concerning the absorbers tested in both campaigns, it must be noted that the performance advantage of the random foam ( $F_s$ ) compared to the tetrakaidecahedron ( $T_s$ ) lattice seen in the first campaign is significantly reduced in the second one. To be clear, both absorbers are the same ones used during the first test



**Fig. 13.** Tetrakaidecahedron ( $T_s$ ), Voronoi ( $V_s$ ), **graded cell size Voronoi ( $V_g$ )**, random foam ( $F_s$ ) and Honeycomb (H) absorber results: (a) absorber thermal efficiency and air outlet temperature, (b) maximum solid inlet temperature and air outlet temperature.



campaign and, it was observed that while the periodic cellular absorber with tetrakaidecahedron morphology remained structurally stable, the unstructured random foam suffered some structural damage at the interstices, with the partial detachment of some ligaments of the foam, which negatively affected its thermal performance. Moreover, the main objectives of the innovative morphologies are not just to enhance the thermal performance of volumetric absorbers, but to improve the mechanical performance of conventional foams. Concerning mechanical performance, the literature shows that ceramic materials can work at high temperatures, but they have poor erosion resistance when used as foams [73], besides, the duration and effect of foam failure depends heavily on the ligaments' microstructure [74]. The new morphologies were manufactured to address mostly the first objective, the improved thermal performance of volumetric absorbers. However, despite the mechanical analysis of the morphologies being beyond the scope of this research, the experiments of the new configurations show that they remain structurally stable without having ligaments failures.

Fig. 13 (b) shows that the best thermal efficiency is linked to the lowest maximum solid temperature at the absorber inlet, which comes close to the definition of the volumetric effect. Moreover, the two absorbers with the lowest maximum solid temperature at the receiver inlet,  $V_g$  and  $V_c$ , are the ones with the lowest extinction coefficients, fitting the pattern presented previously, while the  $F_s$  and  $T_s$  absorbers reverse the pattern in performance, with  $T_s$  having a lower temperature than  $F_s$ . It is assumed that the partial ligament detachment of the foam, reduced the extinction coefficient and the solid temperature, while reducing the specific surface area, which was its distinguishing feature, and hence, its thermal efficiency.

Although the observed changes in thermal efficiency are small for some of the morphologies, the relative variations in efficiency give information on the main tendencies, and the measurements use the same procedure to assure the repeatability of the tests.

## 5. Conclusions

The current work is the first experimental study about the thermal performance of five novel and innovative morphological configurations for use as volumetric receivers in central receiver systems for solar tower plants. The morphologies are based on single unit cells that are continuously repeated, such as diamond cell, tetrakaidecahedron cell, rotated cube cell, cube shifted cell, and a new Voronoi morphology based on a random pattern but with an improved design technique called the Voronoi tessellation technique. These five new configurations together with reference absorbers, silicon carbide honeycomb absorber and conventional unstructured random foam, were tested in two different tests campaigns.

The analysis of the experimental results showed that for the single cell size lattices (D-T-RC-CS) with similar geometrical and optical properties, the specific surface area is a parameter that correctly predicts absorber performance. Furthermore, lattices having the lowest values of either of the main heat transfer parameters, the extinction coefficient or specific surface area, have the poorest thermal performance. Moreover, the experimental analysis of gradual cell size variation, either increasing or decreasing, is not able to enhance the thermal performance of the single cell size absorbers although it is able to decrease the frontal solid temperature. Comparing the decreasing and the increasing cell size pattern, it is seen that each design favour one of the two main heat transfer mechanisms while penalizing the other to finally obtain similar thermal efficiencies.

Finally, only one morphology, Voronoi, with two different patterns, constant cell size and gradually increasing cell size, is able to

surpass the thermal performance of baseline absorbers, as well as having a lower solid temperature at the absorber frontal surface. This is the first work reporting experimental tests on an absorber with a linear cell size variation in it, as produced by the Voronoi tessellation technique.

## CRediT authorship contribution statement

**Antonio L. Avila-Marin:** Conceptualization, Methodology, Formal analysis, Investigation, Resources, Writing – original draft, Visualization. **Jesus Fernandez-Reche:** Methodology, Resources, Supervision, Project administration. **Sandro Gianella:** Resources, Writing – original draft, Visualization, Project administration. **Luca Ferrari:** Resources, Visualization. **Daniel Sanchez-Señoran:** Resources, Visualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Nomenclature

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| $A$             | Frontal surface area ( $m^2$ )                           |
| $d$             | Diameter (m)                                             |
| $h$             | Convective heat transfer ( $W/(m^2 \cdot K)$ )           |
| $L$             | Length (m)                                               |
| $\dot{m}$       | Mass flow rate (kg/s)                                    |
| $P$             | Power (W)                                                |
| $\bar{P}$       | Average power (W)                                        |
| $S_s$           | Wire surface area ( $m^2$ )                              |
| $T$             | Temperature (K)/(°C)                                     |
| $\bar{T}$       | Average temperature (K)/(°C)                             |
| $V$             | Volume ( $m^3$ )                                         |
| $a_v$           | Specific surface area ( $1/m$ )                          |
| $a$             | Absorption coefficient ( $1/m$ )                         |
| $\sigma_s$      | Scattering coefficient ( $1/m$ )                         |
| $\alpha$        | Absorptivity (–)                                         |
| $\beta$         | Extinction coefficient ( $1/m$ )                         |
| $\varepsilon$   | Emissivity (–)                                           |
| $c_p$           | Heat capacity ( $J/(kg \cdot K)$ )                       |
| $\sigma$        | Stefan-Boltzmann constant number ( $W/(m^2 \cdot K^4)$ ) |
| $\eta$          | Efficiency (%)                                           |
| $\varnothing$   | Porosity (–)                                             |
| $\varphi$       | Flux ( $W/m^2$ )                                         |
| $\bar{\varphi}$ | Average flux ( $W/m^2$ )                                 |

## Abbreviations

|      |                 |
|------|-----------------|
| SiC  | Silicon carbide |
| $Nu$ | Nusselt number  |
| $Ra$ | Rayleigh number |
| $Pr$ | Prandtl number  |
| $TL$ | Thermal losses  |
| HE   | Heat exchanger  |

## Subscripts

|      |                |
|------|----------------|
| a    | Air            |
| abs  | Absorber       |
| amb  | Ambient        |
| bk   | Blanket        |
| in   | Inlet          |
| conv | Convection     |
| ext  | External       |
| HE   | Heat exchanger |
| out  | Outlet         |
| rad  | Radiation      |
| rec  | Receiver       |
| s    | Solid          |
| tot  | Total          |
| w    | Water          |
| x    | X direction    |
| y    | Y direction    |

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